

Operating System Notes

Unit 1: Introduction to Operating Systems

An Operating System (OS) is system software that acts as an interface between the user and computer hardware. It manages hardware resources and provides services to application programs.

Functions: Process Management, Memory Management, File Management, Device Management, Security and Protection, User Interface.

Types: Batch OS, Multiprogramming OS, Multitasking OS, Time Sharing OS, Distributed OS, Real-Time OS.

Unit 2: Process Management

A process is a program in execution.

Process States: New, Ready, Running, Waiting, Terminated.

PCB contains Process ID, Process State, Program Counter, CPU Registers, Scheduling Information, and Memory Management Information.

CPU Scheduling: FCFS, SJF, Round Robin.

Unit 3: Synchronization and Deadlocks

Critical Section requirements: Mutual Exclusion, Progress, Bounded Waiting.

Semaphore operations: wait() and signal().

Deadlock is a situation where processes wait indefinitely for resources held by each other.

Deadlock Conditions: Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.

Deadlock Avoidance commonly uses Banker's Algorithm.

Unit 4: Memory Management

Paging divides logical memory into pages and physical memory into frames.

Segmentation divides memory into logical segments such as code, data, and stack.

Virtual Memory allows execution of programs larger than physical memory.

Page Replacement Algorithms: FIFO, LRU, Optimal.

Unit 5: File System

File Allocation Methods: Contiguous, Linked, Indexed Allocation.

Directory Structures: Single-Level, Two-Level, Tree-Structured.

Disk Scheduling Algorithms: FCFS, SSTF, SCAN, C-SCAN.

Frequently Asked Questions

What is a Process? A process is a program in execution.

What is a Thread? A thread is the smallest unit of CPU execution within a process.

What is Deadlock? A situation where processes wait indefinitely for resources held by one another.

What is Paging? A memory management technique that divides memory into pages and frames.

What is Virtual Memory? A technique allowing programs larger than physical memory to run.